

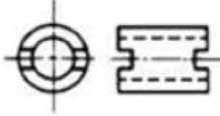
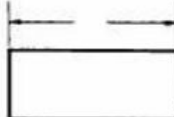
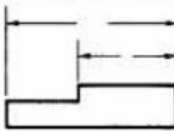


ME 142- ENGINEERING DRAWING & GRAPHICS

TYPES OF LINES

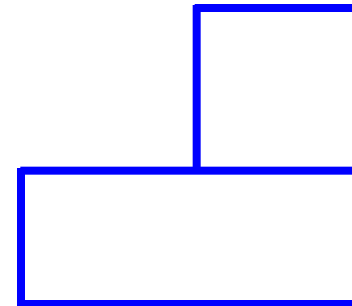
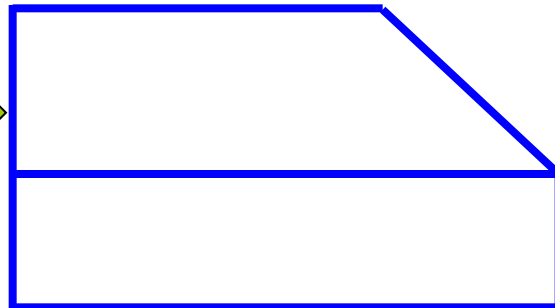
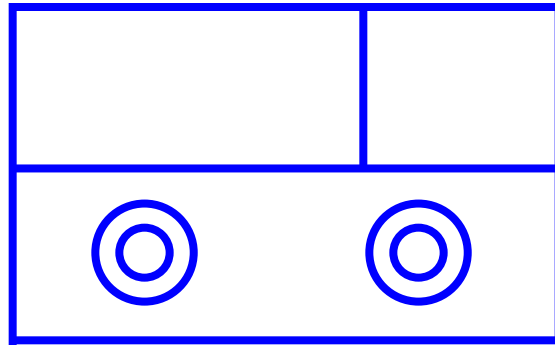
**HITEC UNIVERSITY TAXILA
Department of Mechanical Engineering**

Line Conventions

LINE STANDARDS			
NAME	CONVENTION	DESCRIPTION AND APPLICATION	SAMPLE
CENTER LINES		THIN LINES MADE UP OF LONG AND SHORT DASHES ALTERNATELY SPACED AND CONSISTENT IN LENGTH USED TO INDICATE SYMMETRY ABOUT AN AXIS AND LOCATION OF CENTERS	
VISIBLE LINES		HEAVY UNBROKEN LINES USED TO INDICATE VISIBLE EDGES OF AN OBJECT	
HIDDEN LINES		MEDIUM LINES WITH SHORT EVENLY SPACED DASHES USED TO INDICATE CONCEALED EDGES	
EXTENSION LINES		THIN UNBROKEN LINES USED TO INDICATE EXTENT OF DIMENSIONS	
DIMENSION LINES		THIN LINES TERMINATED WITH ARROWS HEADS AT EACH END USED TO INDICATE DISTANCE MEASURED	

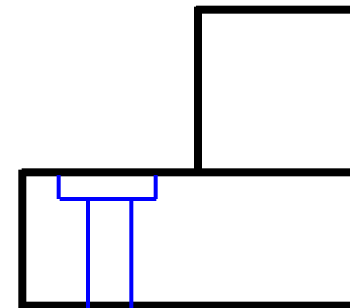
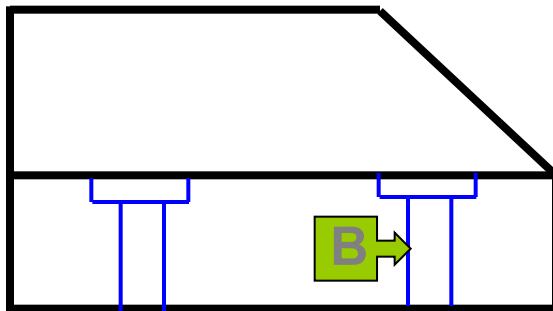
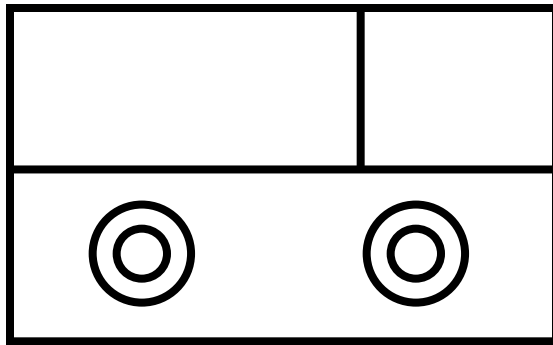
In a technical drawing, each type of line has its own significance. Each is distinguished from the other by a particular thickness or shape.

Object Line



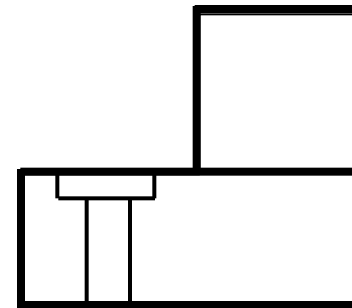
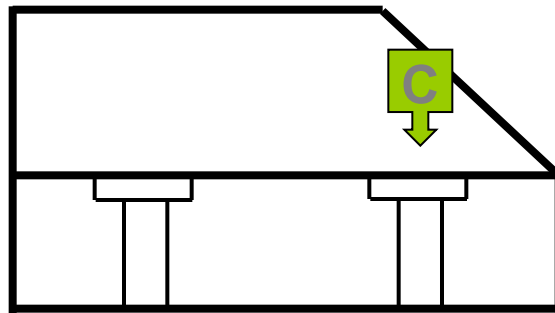
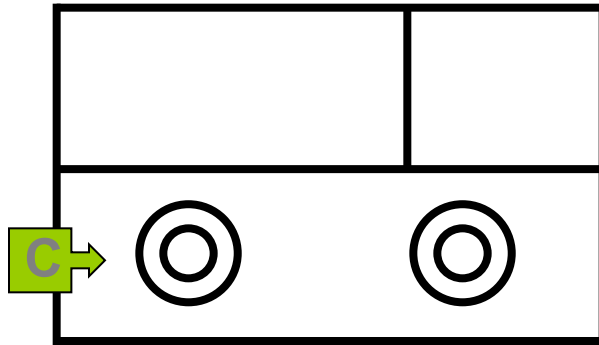
- **The object line or visible outline** is made of a continuous line that portrays the visible forms of the object.
- It is a thick line, its thickness chosen to clearly show the shape.

Hidden Line



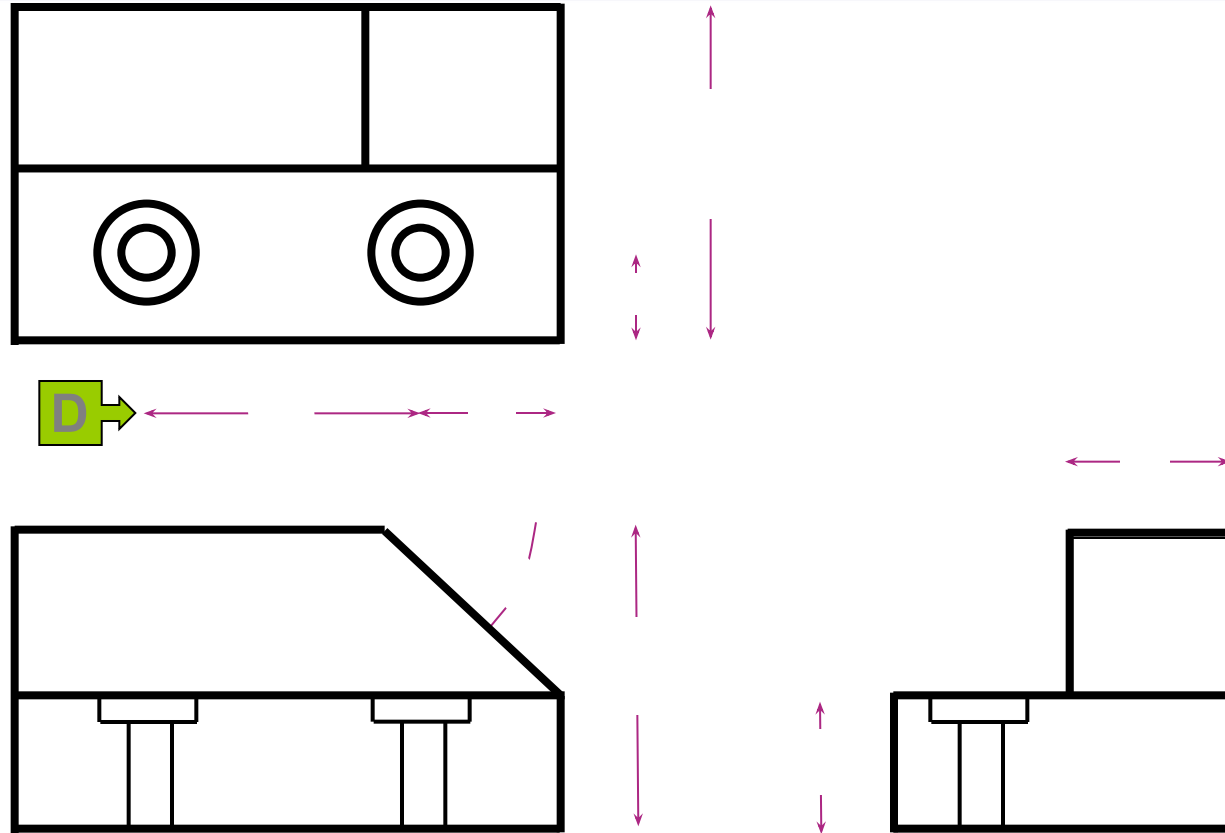
The hidden feature line is made of a dashed line that portrays surfaces and edges that are hidden from view..

Centre Line



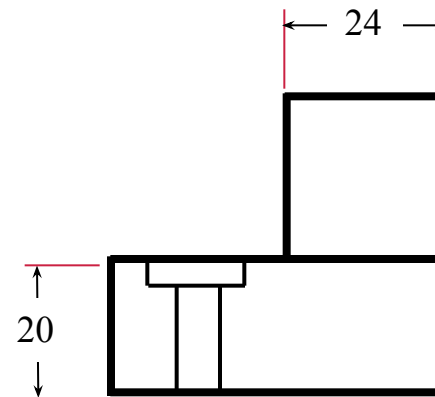
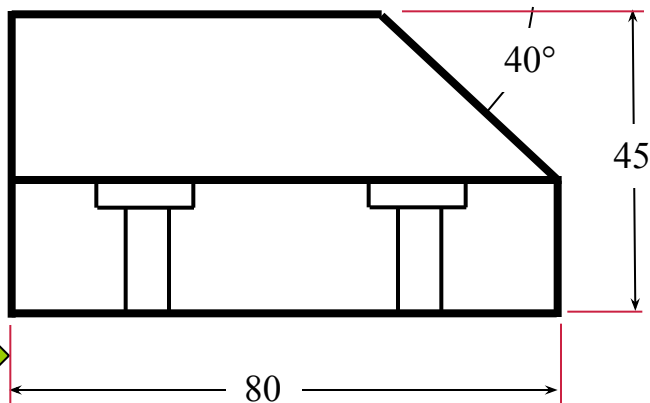
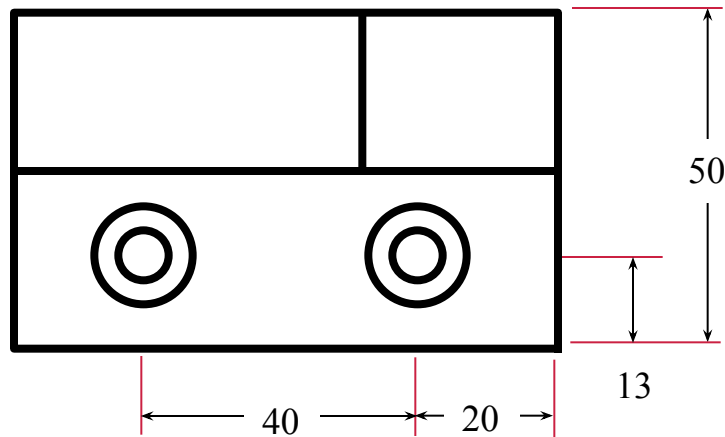
The center line is made from phantom lines. It is used to designate the axes of round or circular shaped objects.

Dimension Line



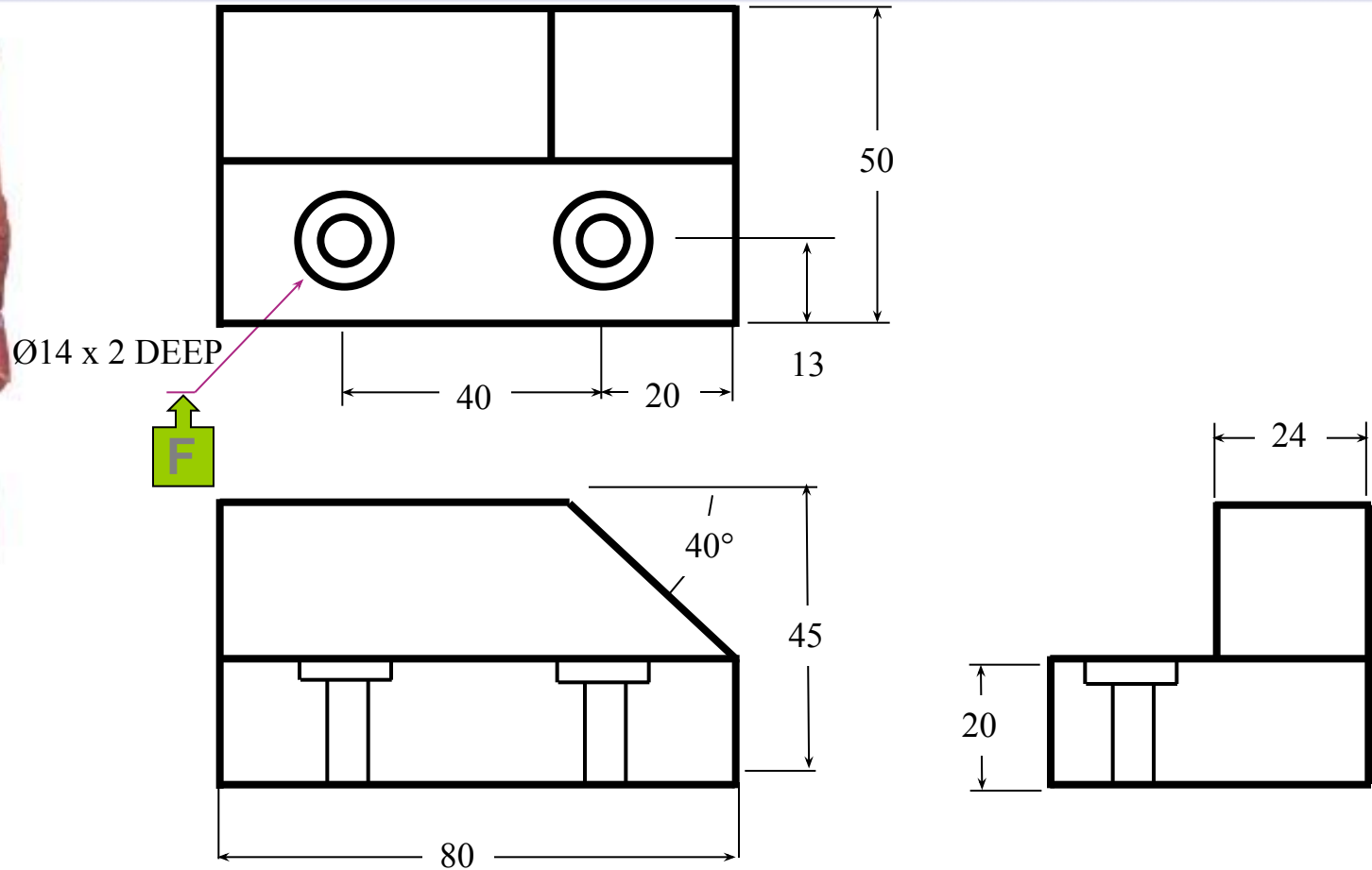
The dimension line is a fine line ending with an arrow. Its purpose is to enter and specify the placement of the dimensions.

Extension Line



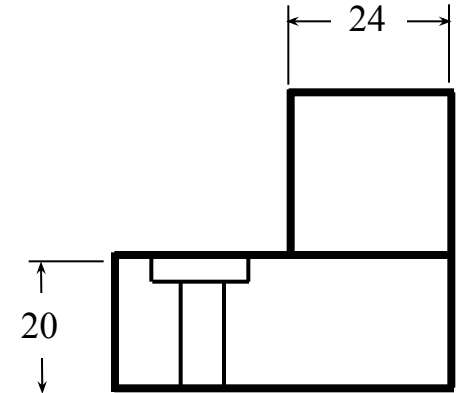
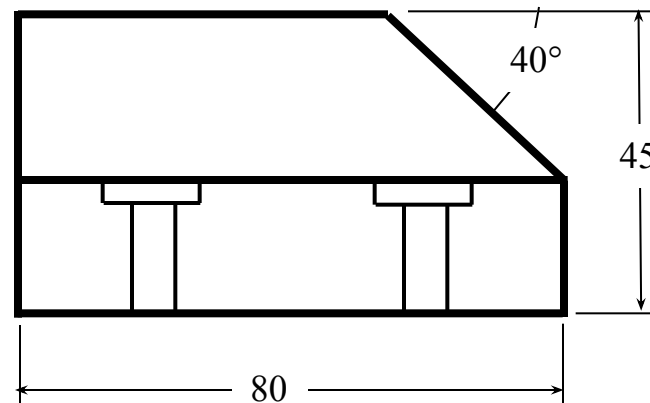
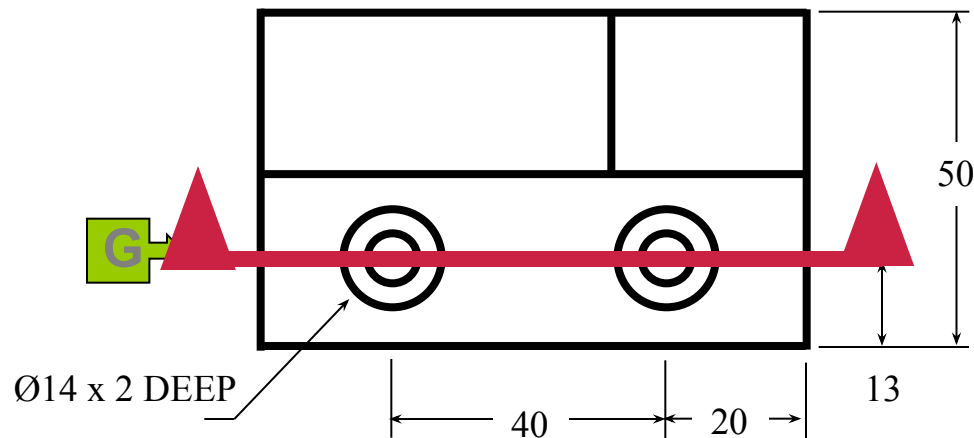
Extension lines are fine lines that indicate the placement of the dimensions.

Leader



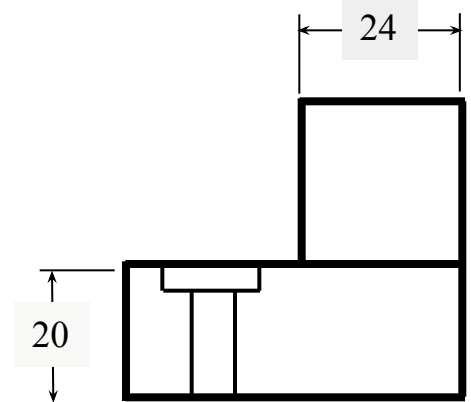
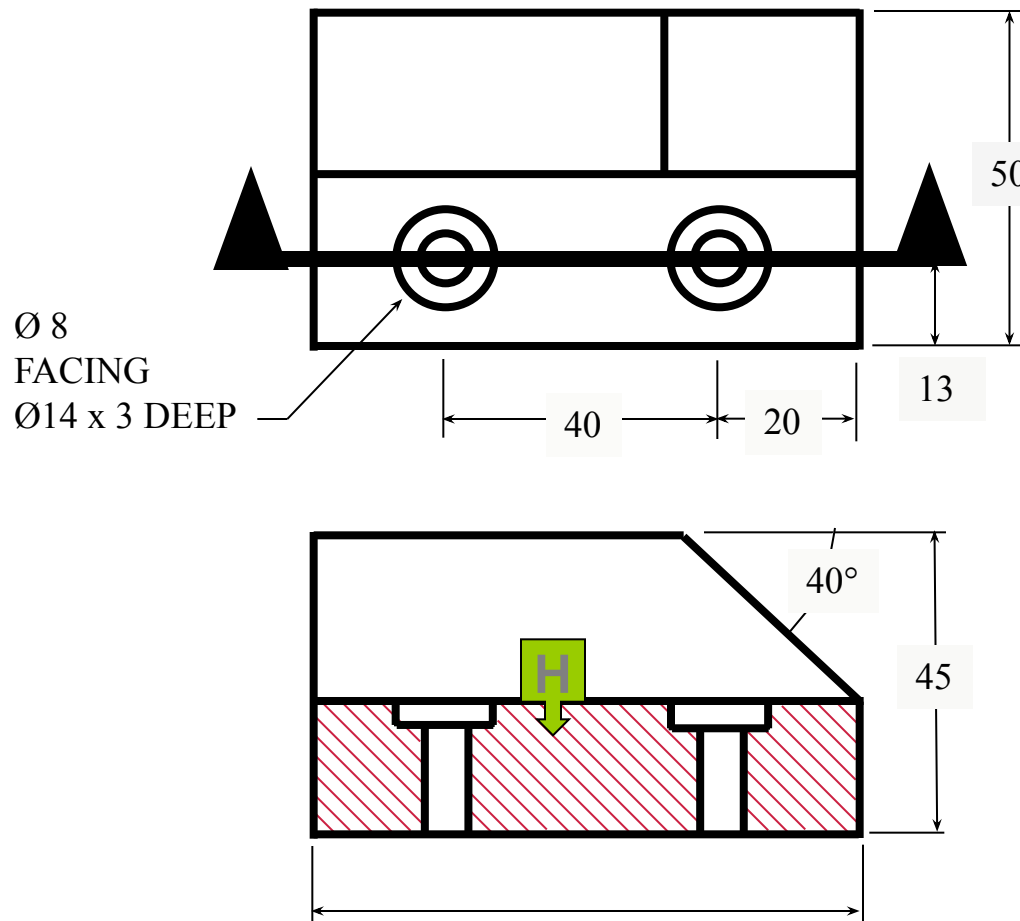
The leader is used to indicate the part of the drawing to which a note refers

Cutting Plane Line



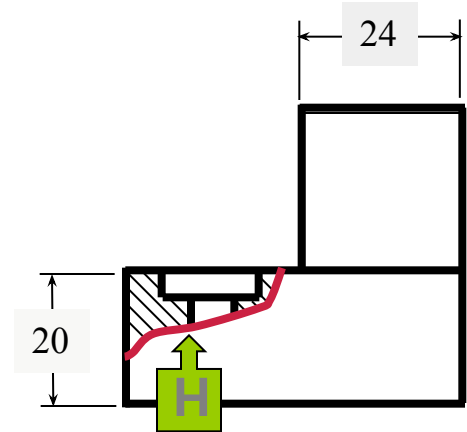
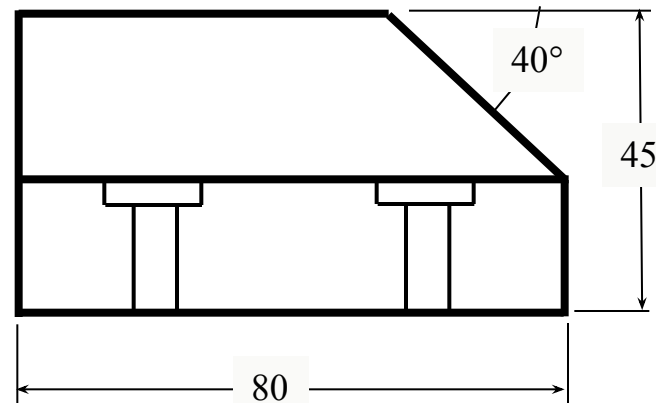
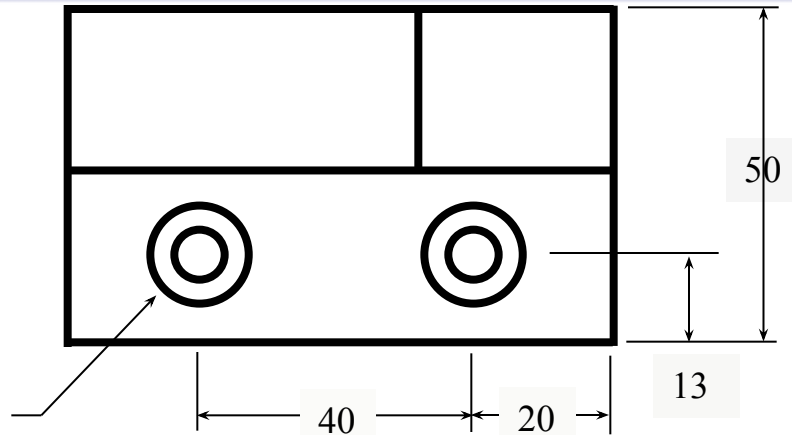
The cutting plane line is very thick; this line is used to designate where an imaginary cut took place.

Section Lines



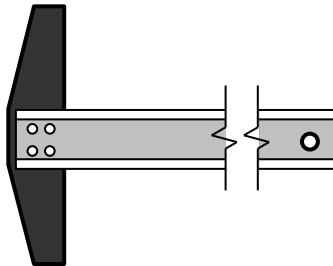
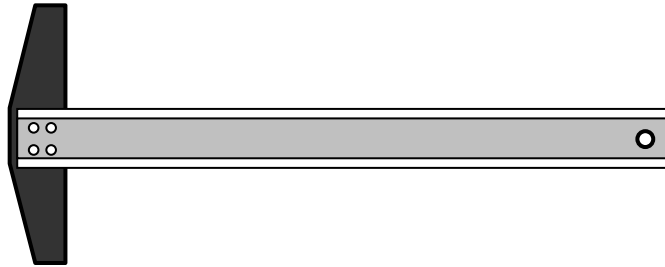
Sections lines are fine lines used to indicate the surface imagined to have been cut

Short Break Line



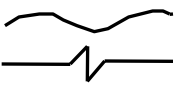
The short break line is a thick, irregular line used to indicate a partial cut. This line is drawn freehand

Long Break Line



The long break line is a fine line used to shorten the view of a long part that can not be shown full length.

Summary (Lines)

	LINE	NAME	PURPOSE	THICKNESS	
A		Object line or visible outline	Portrays visible forms.	Thick	100
B		Hidden feature line	Portrays surfaces and edges that are hidden from view	Medium	50
C		Center line	Portrays the center of a round or circular object.	Fine	25
D		Extension line	Is used to indicate the dimension.	Fine	
E		Dimension line	Is used to indicate the dimension.	Fine	
F		Cutting plane line	Indicates the placement of an imaginary cut.	Very thick	
G		Section lines	Indicate the surface imagined to have been cut.	Fine	
H		Break line	Used to shorten the view of a part.	Fine	
J		Leader	Indicates the part of a drawing to which a note refers	Fine	

Line Conventions (Contd.)

